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ORIGINAL ARTICLE

Media analytics via machine learning: social media engagement prediction for TV channels

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Abstract
Analyzing social media data has become increasingly important for TV channels as it offers valuable insights into audience preferences, engagement, and sentiment. It provides the feedback needed to align content with audience tastes, fostering greater viewer satisfaction and loyalty. In this paper, we propose a media analytics framework for analyzing audience engagement with posts made by TV channels on their social media platforms. We combine traditional statistical modeling, natural language processing tools, and explainable machine learning to predict engagement and derive feature importance. Our results on two platforms (Facebook and Instagram) for a major TV channel in Chile confirm the predictive capabilities of machine learning: Extreme Gradient Boosting achieved the best performance on Facebook, with a Mean Absolute Percentage Error (MAPE) of 25.14%, while the lowest MAPE for Instagram (16.56%) was obtained using Support Vector Regression. Furthermore, explainable machine learning techniques unveil interesting conclusions for decision-making, such as the importance of mentions of TV personalities, which can be automated using Named Entity Recognition.

Keywords Media analytics · Social media · Explainable artificial intelligence · Engagement prediction

1 Introduction

The television industry has evolved significantly due to technological diversification. To maintain competitiveness, channels increasingly rely on large datasets (including ratings, audiovisual content, and social network data) to drive analytical business decisions (Dodds et al. 2023).

Social networks have established themselves as essential tools for mass communication and promotion (Rodríguez-Ibánez et al. 2023). These platforms offer real-time feedback that informs programming and marketing (Porter et al. 2023), while simultaneously allowing channels to track demographics and measure campaign effectiveness (see, e.g., Amalvy et al. 2024; Malik and Shakshuki 2022).

However, a critical gap remains in understanding how content drives user engagement (Dolan et al. 2019; Elhussein 2025). Failing to measure these effects hinders editorial decision-making and limits the application of media economics. Accurate prediction is necessary to move audience analytics from descriptive reporting to proactive, strategic planning (Drivas et al. 2022; Elhussein 2025).

To address this limitation, we propose a novel AI-powered system for automatic social media analysis. By utilizing techniques such as machine learning and Natural Language Processing (NLP), this approach identifies patterns and derives valuable insights to support decision-making (see, e.g., Porter et al. 2023).

While AI-powered systems have been used for analyzing media data (see, e.g., Mondragon et al. 2018; Yin et al. 2023), most applications focus on descriptive analytics, such as viewership prediction or sentiment analysis, rather than providing forward-looking, prescriptive guidance for content creation. Media companies still struggle to understand

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the specific drivers of user engagement and how to systematically design content to maximize audience interaction. To address this gap, this study explores the following research questions:

(RQ1) Can we predict engagement TV channels' social media accurately to tailor content to audience preferences?
(RQ2) Does the mention of relevant personalities in TV programs influence audience engagement?
(RQ3) Are NLP variables useful for predicting audience engagement?

Furthermore, our study discusses the following aspects that are relevant and novel for the state of the art in media analytics:

- In order to address all the research questions, we propose a framework based on predictive analytics that allows the evaluation of different data sources. We show empirically how the combination of sources improves performance in the context of regression. Boruta-SHAP, a well-known wrapper feature selection method from the eXplainable Artificial Intelligence (XAI) is used to identify relevant variables and depict their contribution.
- We discuss various approaches for feature engineering in order to design relevant covariates based on the content of the post. We extract relevant social media features such as hashtags and emojis, and specific information such as mentions of TV personalities related to programs. This process is related to RQ2.
- In order to address RQ3, we propose utilizing NLP techniques such as Named Entity Recognition (NER), polarity detection, and sentiment analysis or automatic feature engineering. The goal is to complement manually-created features and other information that can be extracted from posts.
- We collect a rich dataset from a Chilean TV channel, consisting of 3556 posts made by the channel in two social media platforms, Facebook and Instagram, related to morning talk show. This dataset allowed us to validate our framework and visualize relevant patterns for enhancing the design of posts and improving the TV program.

The organization of this paper is as follows: First, Sect. 2 reviews prior studies on engagement on social media, discussing also AI-powered decision systems for TV channels. Next, in Sect. 3, we present the proposed framework for social media engagement prediction for TV channels. Then, we report experiments conducted on proprietary datasets from a Chilean TV channel in Sect. 4. Subsequently, Sect. 5 discusses managerial implications, offering operational strategies for platform-specific optimization and synchronizing social content with broadcast peaks. Finally, Sect. 6 outlines the main findings and suggests directions for future research.

2 Prior work on social media engagement

Social media engagement encompasses user interactions such as reactions, shares, and comments (Balaji et al. 2023; Sadeque et al. 2019). While substantial research focuses on user-centric models (predicting churn or future participation levels based on observation windows and activity thresholds (Dror et al. 2012; Sadeque et al. 2015; Danescu-Niculescu-Mizil et al. 2013)) our study adopts a post-centric approach. We analyze specific TV channel posts to predict success rather than user retention. A similar post-based strategy was employed by Dai and Wang (2021) using multimodal BERT; however, such "black-box" methods often obscure the specific drivers of engagement.

Feature engineering is critical for predicting post success. Linguistic elements, such as emotionality and complexity, directly influence interaction (Deng et al. 2021; Babu and Kanaga 2022; Kabir et al. 2018), as does the specific subject matter identified through topic mining (Abkenar et al. 2021). Structural and formatting choices are also pivotal: while longer posts provide information, they may induce boredom, and "shouting" (all-caps) can denote urgency or excitement (Hamilton et al. 2017; Arguello et al. 2006; Pino et al. 2019). Furthermore, visual content, URLs, and hashtags tend to stimulate engagement (Aydin 2020; Adams 2016; Sedhai and Sun 2014; Taecharungroj 2017), while interactive strategies like questions (Lei et al. 2017) and "Best Time to Post" (BTTP) variables (e.g., seasonality, day of the week) significantly impact performance (Mariani et al. 2016; Dai and Wang 2021).

A significant branch of recent research focuses on Multimodal Learning. Works by Hu et al. (2016) and others (see, e.g., Kruk et al. 2019; Xu et al. 2020) show that fusing textual embeddings (e.g., BERT) with visual embeddings (e.g., ResNet, VGG) significantly outperforms text-only models. For instance, determining whether an image matches its caption's intent can improve engagement prediction accuracy by nearly 10% (Kruk et al. 2019).

Modeling engagement requires navigating trade-offs between accuracy and interpretability. Logistic regression offers transparency (Danescu-Niculescu-Mizil et al. 2013), while Support Vector Machine, Random Forest, and Deep Learning techniques often yield higher predictive accuracy at the cost of explainability (Mahmud et al. 2014; Hamilton et al. 2017; Rettinger et al. 2025; Dai and Wang 2021; Brito

and Adeodato 2023). These models must also account for platform-specific contexts (such as the visual focus of Instagram versus the textual nature of Facebook) and user-based factors like demographics and acculturation (Bharanitharan et al. 2024; Sadeque et al. 2019; Lai et al. 2022).

In the television industry, AI and sentiment analysis are increasingly used to shape editorial decisions and predict viewership (Anantrasirichai and Bull 2022; Mondragon et al. 2018; García-Perdomo 2021; Cheng et al. 2016; Min et al. 2015; Malik and Shakshuki 2022; Kumar et al. 2017). However, current literature rarely balances predictive power with interpretability or utilizes industry-specific variables, such as the presence of TV personalities. Our framework addresses these gaps by evaluating both transparent and complex models to not only assess post potential but also identify the specific drivers of success.

3 Proposed framework for social media engagement prediction for TV channels

In this section, we formalize a framework based on predictive analytics to extract knowledge from social media data for TV channels. Our objective is to combine two types of variables: manually-designed attributes tailored for the task of post success prediction in terms of viewer engagement, and features obtained automatically via NLP techniques. From a predictive perspective, the combination of these two sources addresses RQ1 (Can we predict TV channels' social media engagement accurately to tailor content to audience preferences?). From an explainability perspective, we aim to answer RQ2 (Does the mention of relevant personalities in TV programs influence audience engagement?) with the first group of variables, while the second group relates to RQ3 (Are NLP variables useful for predicting audience engagement?).

The proposed framework is a customization of well-known processes for knowledge discovery in databases, such as CRISP-DM (Chapman et al. 2000). It also adapts recent AI-based systems based on social media and other textual data (see, e.g., Kodati and Dasari 2024; Saumya et al. 2024) or that combine different data sources (see, e.g., Sánchez et al. 2022) to the task at hand.

Step 1 consists of data preparation, the definition of predictive tasks, and feature engineering, where we design variables relevant to the given task. In Step 2, the best-performing regression model is selected by exploring the predictive performance of different techniques. Step 3 involves model evaluation and interpretation of the mined patterns. For this task, we consider Boruta-SHAP for feature ranking and selection to improve predictive performance and gain a better understanding of the process that generated the data. Finally, Step 4 discusses the decision-making process behind social media analysis in TV channels, which can benefit from the novel framework presented in this paper.

3.1 Step 1: Task definition, feature engineering, and data preprocessing

The initial step involves defining the predictive task and engineering a comprehensive set of features from the raw social media data. The task is cast as a regression problem where the goal is to predict a post's engagement level. To provide a concrete understanding of the source data, the following are anonymized examples of posts from the dataset:

- Example 1 (Facebook Photo): "Longtime member of #[Morning Show Name] bids farewell to the team and moves the panel [emoji:sad-but-relieved-face]: We wish you the best".
- Example 2 (Instagram Reel): "How flirty! Today on #[Morning Show Name] we were with [Band Name], an occasion where [Band's Vocalist Name] took the opportunity to compliment our [Presenter-F] [emoji:face-with-hand-over-mouth] #[TV Channel Name]".

To predict the success of various posts made by TV channels, we propose casting the task of engagement prediction into a regression problem. As done in Coffey and Hollifield (2023), we first define engagement as the sum of all possible interactions with the post, including reactions, comments, shares, clicks on hyperlinks, etcetera. We limit our study to two different platforms: Facebook and Instagram. For the former, we consider reactions, comments, shares, and clicks. For the latter, only reactions and comments are counted.

There is, however, an important flaw in assuming that every interaction is created equal. This simplistic approach masks crucial nuances in user behavior and intent. A reaction is a low-friction, passive acknowledgment. It requires minimal cognitive or behavioral effort and often signifies simple agreement or fleeting appreciation. A comment, conversely, demands active participation. The user must pause, formulate a thought, and invest the time to contribute to the conversation. Clicks are also important to drive driving traffic to external websites, such as publishers. A click on a link within a post is a clear signal of intent to learn more and a critical step in the conversion funnel. Finally, Shares represent the highest tier of engagement: an act of public endorsement. When a user shares a piece of content, they are leveraging their own social capital and reputation to amplify the brand's message to their personal network. This action transforms them from a passive consumer into an active brand advocate (Lazzaro 2024).

In order to take this analysis into account, we propose a variation of the Interactivity Index (InI), developed by the social media analytics firm Sotrender,[1] which is one of the most well-documented weighted engagement models. This strategy, however, dos not consider clicks. The specific weights we assign to the various engagement forms are as follows: Likes (or any other reaction)= 1, Clicks = 3, Comments = 5, Shares = 8.

Post engagement tends to follow a skewed distribution: most posts receive few interactions, with a large number of comments and likes in exceptional cases. To deal with this issue, we suggest applying the Box-Cox transformation to the target variable, also known as logarithmic transformation ($Y := ln(Y + 1)$). Thanks to this transformation, we did not apply outlier removal in terms of engagement. However, we did apply one-hot encoding for categorical features.

We posit that the "black box" of social media engagement can be unlocked by combining three distinct streams of data:

- *Structural* (S) features: The metadata of the post (e.g., format, length, punctuation).
- *Tailored* (T) features: Domain-specific variables manually curated for the TV context (e.g., the specific presence of show hosts, the TV rating of the preceding broadcast).
- *NLP*-Derived features: Automated insights extracted from the text using deep learning models (e.g., Sentiment, Named Entity Recognition).

We constructed a rich feature space derived from three distinct sources. This multi-view approach allows us to isolate the impact of editorial choices (Structural), talent decisions (Tailored), and automated content analysis (NLP). The feature contribution can be assessed in a supervised manner, allowing an effective evaluation of the relevant variables. The different sets of suggested variables are described next. The detailed taxonomy of the feature engineering framework is provided in Table 5; see Appendix A.

- *S.Type*: The nature of the publication can lead to different viewer engagement. Among the possible post types, we consider photos, stories, or reels depending on the social media platform.
- *S.Content*: Dichotomous variables that describe the content of the post can be created. Some examples are whether the publication has hashtags, emojis, mentions, question or exclamation marks, or upper-cased words.
- *S.Length*: The length of the post can be used as a covariate. This can be a numerical variable, or a categorical variable that indicates a post is long by defining a threshold in terms of the number of characters.
- *T.Personalities*: TV personalities related to a specific program, such as the presenters, can be mentioned in the post. This is a crucial analysis for TV channels because they can replace presenters and other roles based on popularity. Social media data is a cheap and valuable source of information, in contrast to surveys, for example.
- *T.Rating*: Posts can be aligned with the TV rating (percentage of viewers) of the program. A good time to post can be after a successful program in terms of audience. The average TV rating of the program before posting can be considered an explanatory variable.
- *T.BTTP*: The customer service literature defines "best time to call" (BTTC) variables that allow the identification of the best day and time to call a customer (see, e.g., Sánchez et al. 2022). Similarly, we consider BTTP variables reflecting the time of day (morning or evening) or the day of the week, as suggested in Dai and Wang (2021).
- *NLP.Sentiment*: We can use sentiment analysis to label each post in terms of its polarity (positive, neutral, or negative). As a dichotomous variable with three levels, one-hot encoding should be used, dropping one of the categories to avoid multicollinearity.
- *NLP.NER*: Similarly, we can utilize NER to identify whether the name of any person is mentioned in the post. Note that this is different from mentions from the S.Content category (@mentions), which refer to tags to persons but also brands or other programs, among others.

Regarding the construction of the NLP variables, we consider a transition-based NER approach to identify named entities (names of people, in particular) in text using a sequence of actions or transitions. These transitions guide the model through the text, deciding how to label each token in the sequence. A pre-trained large language model (LLM) is considered for this purpose. For sentiment analysis, we consider pre-trained transformer-based models to label each post as positive, neutral, or negative. For both models, there is no need to fine-tune them as they can be applied directly to new data.

After the features are computed, data preprocessing must be performed. Note that tabular and textual data require different techniques. For the former, outlier removal, missing data imputation, one-hot encoding, and feature selection are suggested (Sánchez et al. 2022). For the latter, we consider stemming, which involves removing the last few characters

[1] What is the Interactivity Index? Sotrender Blog, accessed: october 23, 2025, www.sotrender.com/blog/faq/what-is-the-interactivity-index/

from a word, lemmatization, which involves converting a word into its meaningful base form, and tokenization, where information such as dates or phone numbers is replaced by a surrogate value to be recognized by the language model (Vairetti et al. 2024).

3.2 Step 2: The learning process

It is well-known that no machine learning method consistently outperforms others in terms of predictive performance when dealing with tabular datasets. Therefore, we must evaluate a wide variety of models using an adequate validation setting (Sánchez et al. 2022). This is also confirmed in the literature on social media engagement prediction (see Sect. 2).

In this study, we explore the performance of various linear methods, tree-based approaches, ensembles, and other traditional machine learning techniques. These techniques are described next, with a detailed description for each model available in Hastie et al. (2009).

- *Multiple Linear Regression:* This model relies on the linear relationship between the independent and dependent variables and is trained using Ordinary Least Squares (OLS). The main advantage of this strategy is that hypothesis tests can determine the statistical significance of the predictor variables. This helps assess whether a particular variable contributes significantly to the model. Formally, a linear model has the following form:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon \tag{1}$$

where Y is the target variable, X_j are the covariates, β_j are the coefficients, and ϵ is the error term, with $j = 1, ..., p$. For each coefficient, the t-statistic is computed as $t_j = \frac{\beta_j}{SE(\beta_j)}$, with $SE(\beta_j)$ being the standard error of the estimated coefficient. A backward elimination approach for feature selection is suggested, discarding non-significant variables (p-value larger than 0.05; the t-statistic is determined using the t-distribution with $n - p - 1$ degrees of freedom, where n is the sample size and p is the number of predictors) one by one until all the variables in the model are statistically significant.

- *Ridge Regression:* This approach is similar to OLS regression, but the optimization process includes a regularization term called ridge or l_2 norm. The goal is to reduce the risk of overfitting by minimizing the sum of the squared values of the coefficients. This approach requires tuning a hyperparameter λ that balances the trade-off between regularization and model fit, as follows:

$$Loss_{ridge} = \sum_{i=1}^{n} (y_i - \mathrm{x}_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \tag{2}$$

- *LASSO Regression:* This approach is similar to ridge regression in the sense that it is a linear model that includes a regularization term. Instead of using the squared values of the coefficients, the LASSO penalty minimizes the absolute values (l_1 norm). This strategy is useful for feature selection, retaining only those variables that contribute to the prediction of social media engagement.
- *Decision Trees:* These models use tree structures to make decisions, dividing the input data space in ways that minimize an error criterion. All created variables were initially included. Decision trees select variables based on their ability to reduce the mean squared error (MSE).
- *Random Forest:* This ensemble of tree-based regression models utilizes bagging (sampling with replacement) to create multiple different subsets. At each split in the tree, a random subset of features is considered. The final prediction is obtained by averaging the predictions of all individual trees.
- *Extreme Gradient Boosting (XGBoost):* This is an adaptation of gradient boosting (a tree-based ensemble approach in which learners are built sequentially, with each new tree trained to correct the errors of the previous ones) that introduces additional enhancements to improve efficiency and performance. XGBoost includes parallel processing, resulting in faster training times. Additionally, it incorporates LASSO or ridge regularization to prevent overfitting and improve model generalization.
- *K-Nearest Neighbors:* This is a traditional machine learning technique that predicts the target value of a new data point based on the average of the target values of its k nearest neighbors in the feature space. The algorithm requires a distance metric to measure the similarity between data points.
- *Multi-layer Perceptron (MLP):* A fully-connected neural network with a single hidden layer is also considered a regression model. The main virtue of MLP is its ability to construct non-linear functions, capturing complex data patterns.
- *Support Vector Regression:* This technique also fits a non-linear decision surface considering the hinge loss function together with the l_2. The non-linearity is achieved by the kernel trick, which projects the data points into a higher-dimensional space.

3.3 Step 3: Model evaluation and interpretation of the mined patterns

Model evaluation is a crucial step for selecting the best regression model and defining the optimal hyperparameter configuration. A training/test partition is suggested, with 80% of the data allocated for training the models and the remaining 20% reserved for the final model evaluation. Hyperparameter selection should occur within an inner validation loop using the training set. This approach ensures that the test set remains unseen during the model calibration phase (see, e.g., Sánchez et al. 2022).

Social network analytics is prone to distribution shift, as major changes may occur between the period when the training data is collected and when the predictive model is applied, potentially leading to a significant drop in performance (Liang et al. 2025). In such cases, a common approach is to compute the prequential accuracy using an out-of-time test sample (out-of-time holdout), which has been shown to be more suitable than random holdout in dynamic environments (Liang et al. 2025).

The following performance metrics are suggested for model evaluation in the context of regression:

- *Root Mean Squared Error (RMSE):* It is computed as the square root of the average squared differences between the predicted values $\hat{y}_i$ and the actual values y_i, as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{3}$$

- *Mean Absolute Percentage Error (MAPE):* It is computed as the average absolute percent error for each time period minus actual values divided by actual values, as follows:

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{4}$$

- *R-squared* (R^2): It corresponds to the proportion of the variance for a target variable that is explained by the covariates in the model. In contrast to the error measures RMSE and MAPE, it provides an indication of the goodness of fit and ranges from 0 to 1, where a value closer to 1 indicates a better fit. R^2 is computed as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5}$$

where $\bar{y}$ is the mean of the actual values.

A crucial step in this framework is the use of eXplainable Artificial Intelligence (XAI) to interpret the results and assess the influence of different variables on the model's outcomes to gain deeper insights. Apart from the interpretation we gain from intrinsically interpretable models such as Multiple Linear Regression and Decision Trees, we adopt the BorutaSHAP approach for feature selection and interpretation of machine learning strategies. Recent studies have highlighted its effectiveness across various domains, including operations management and prescriptive analytics (De Bock et al. 2023).

BorutaSHAP integrates Boruta's feature selection capability with SHAP's interpretability to improve model transparency and performance. SHAP computes Shapley values, originating from coalitional game theory, to estimate the contributions of individual attributes. These values disaggregate the model performance into contributions from each attribute, akin to distributing payouts among "players", which are variables in the model (Sánchez et al. 2022).

While computing exact Shapley values can be computationally intensive, BorutaSHAP offers an efficient heuristic for assessing feature importance. It also facilitates the creation of visually compelling representations of variable contributions. Initially, Boruta identifies a subset of relevant features. This strategy compares the importance of each variable with that of randomly permuted copies of itself. Variables that consistently have higher importance than their permuted counterparts are deemed significant. Then, SHAP values are computed for these selected features to understand their impact on model predictions.

BorutaSHAP also serves as a feature ranking method, evaluating the significance of each variable within a black-box model. This ranking can guide a backward elimination process, where the top $p^* \leq p$ most influential variables are selected, and the model is retrained on different subsets of variables. The choice of p^* is user-defined based on variable importance, with BorutaSHAP plots aiding in this decision-making process (Sánchez et al. 2022).

3.4 Step 4: The decision-making process

The final phase of the model involves the decision-making process, which can be either short-term (operational) or medium-term (tactical). In the short-term case, the predictive models generate probabilities for posts' success in terms of engagement, which can be used to design more effective interactions with TV viewers. A simulator can be created to score possible posts and predict their engagement, allowing for changes in structure and content. Additionally, the insights gained from the XAI approaches discussed in Step 3 can be utilized by the community managers at the

TV channel. Here, the most relevant variables are structural and NLP features.

In terms of tactical decision-making, the TV channel can utilize this framework to make adjustments to TV programs. For example, the engagement resulting from mentioning TV personalities can be analyzed to determine the optimal time for adjustments, such as changing hosts, reporters, or panelists. Alternatively, variables related to different sections of a specific program can be created to evaluate audience perception when mentioned in posts. These decisions can be made by correlating TV ratings with viewer engagement, as suggested in Ueoka and Ishii (2018). In contrast to operational decisions, the most relevant variables are tailored features specifically designed for evaluating these tactical decisions.

The framework can be extended to user-based engagement prediction to better understand viewer preferences. This can be particularly useful for online platforms where content can be personalized via recommender systems (see, e.g., Yin et al. 2023). A user-based approach can also help mitigate data issues. For example, to account for "heavy users" who like all posts, a more sophisticated definition of engagement based on user behavior can be designed.

However, there are some issues with social media data that need to be considered when applying this framework. For example, fractured consumption of multiplatform TV refers to the way audiences consume television content across various platforms and devices, leading to fragmented viewing patterns. This poses challenges for content creators in reaching and engaging their target audiences. Furthermore, the audience is spread across different platforms and devices, making it harder to capture a large, unified audience for a single broadcast or show (Coffey and Hollifield 2023). For instance, social media users, especially on Instagram, are likely to be younger and may consume TV online, thus not being counted in traditional TV rating measures. This phenomenon leads to biases when measuring and making decisions based on both TV ratings and social media engagement. Additionally, measuring visits could also generate numerical distortions, as people do not always browse from the same device. A person who connects from their home computer, then from their cellphone, and later from their office computer would be counted as three different unique visitors, despite being a single person. Therefore, there is a pressing need for integrated systems that can identify users across multiple platforms (Fulgoni and Lipsman 2017).

4 Experimental results

We implemented the proposed viewer engagement framework based on social media data from a Chilean TV channel. Section 4.1 provides a detailed description of this dataset and the experimental setup. A summary of the results is presented in Sect. 4.2. Finally, Sect. 4.3 offers a summary of the main insights derived from the models and discusses their managerial implications.

4.1 Experimental setting and dataset

The dataset comprises 3556 posts made by the channel in Facebook and Instagram (2925 and 631 samples, respectively), collected from January to December 2022. This 5:1 ratio is a strategic artifact that offers deep insight into media economics. The channel uses Facebook as a high-frequency broadcast tool. Because Facebook allows outbound links, the channel posts dozens of times a day (clips, news snippets, and articles) mainly to drive traffic to their website where ad revenue is generated.[2] In contrast, Instagram is treated as a premium, low-frequency channel. Without the ability to easily link out (except in Stories), the goal is purely brand engagement. Content is curated, visual, and less frequent to avoid algorithmic penalization or follower fatigue. This context is vital for interpreting the results: the Facebook model must find signals in a noisy, high-volume stream, while the Instagram model operates on a cleaner, more curated dataset.[3]

All posts included in the analysis pertain specifically to content related to the channel's morning talk show, providing a focused look at engagement within this popular television genre. While this specific focus allows for a detailed case study, it is important to acknowledge that the audience engaging with this content represents viewers interested in this particular channel and program type. Consequently, the observed engagement patterns, while significant within the national context, may not fully generalize to the entire diverse Chilean television-viewing population or to other program genres.

Figure 1 presents the monthly distribution of posts for the two social networks. From this figure, we observe that the majority of the posts are made for the Facebook platform, with a peak in the months between May and July.

Each record represents a post. We define a numerical target variable that summarizes viewer engagement by adding all the social media interactions for a given entry: reactions, comments, shares, and clicks for Facebook, and reactions and comments for Instagram. The distribution of the post engagement and its logarithmic transformation is presented in Figs. 2 and 3 for Facebook and Instagram, respectively.

[2] Instagram vs Facebook: Which Is Better for Video Marketing? Wyzowl Blog, accessed: october 23, 2025, https://wyzowl.com/instagram-vs-facebook/

[3] How Often To Post on Social Media: Finding Your Perfect Frequency. SocialPilot Blog, accessed: october 23, 2025, https://www.socialpilot.co/blog/how-often-to-post-on-social-media

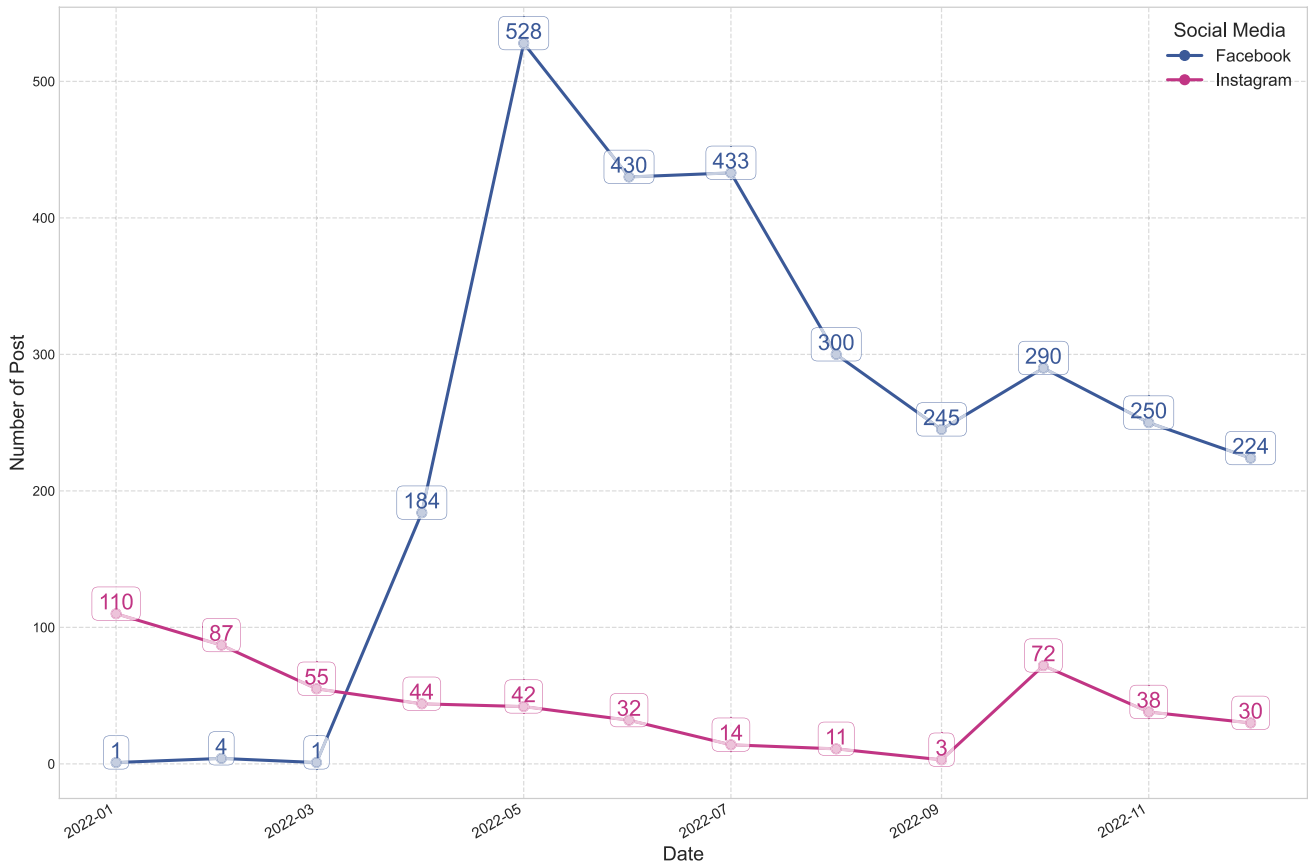
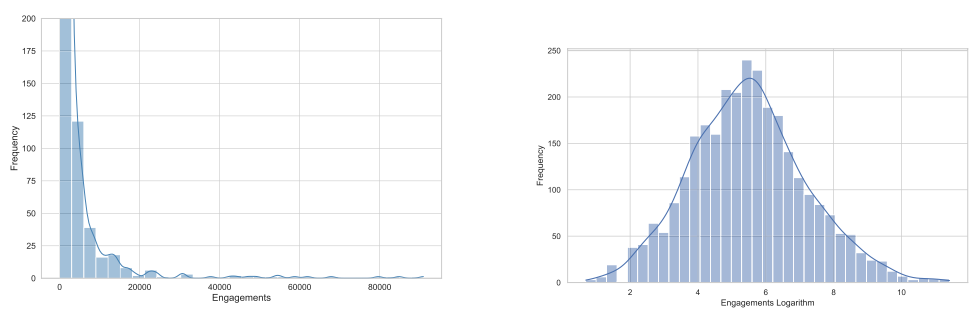


Fig. 1 Post distribution over time for the two social media platforms (Facebook and Instagram)

Fig. 2 Engagement distribution for the Facebook platform. Histograms representing the original distribution and the logarithmic transformation

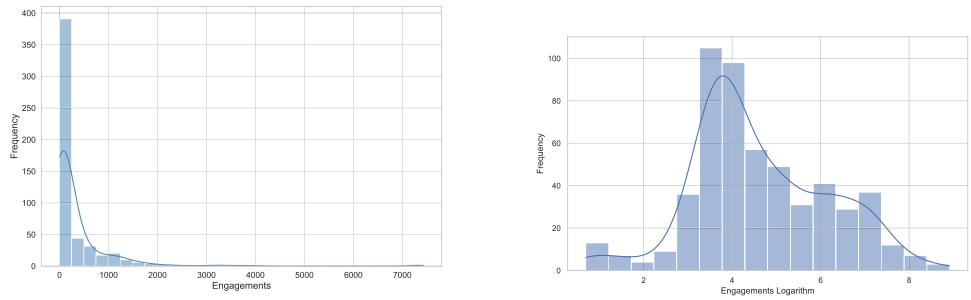


(a) Original distribution (b) Logarithmic transformation

It can be observed in Figs. 2, 3 that the logarithmic transformation results in a more symmetric and bell-shaped engagement distribution, less influenced by outliers. We consider this transformed target variable for the learning task.

Regarding model evaluation and hyperparameter selection, the performance of the models was analyzed using the validation setting discussed in Sect. 3.3. The following sets of hyperparameters were considered:

- *Ridge/LASSO Regression:* We explored different combinations for the trade-off hyperparameter $\lambda \in \{0.1, 1, 10, 100\}$. Note that Multiple Linear Regression does not include any hyperparameters.
- *Decision Trees:* We explored the following values for the maximum depth of the tree and the maximum number of leaf nodes: $\{3, 4, 5\}$ and $\{5, 10, 20, 25\}$, respectively.
- *Random Forest:* For this method, we analyzed the same pruning parameters as for the decision trees. We explored the following values for the number of trees and the number of variables: $\{100, 200, 300\}$ and $\{2, 4, 6\}$, respectively.
- *XGBoost:* Similar to the previous methods, we considered the same values for the maximum depth ($\{3, 4, 5\}$)



(a) Original distribution (b) Logarithmic transformation

Fig. 3 Engagement distribution for the Instagram platform. Histograms representing the original distribution and the logarithmic transformation

and the number of tree-based learners ($\{100, 200, 300\}$). The following values were used for the learning rate: $\{0.01, 0.1, 0.2\}$.

- *K-Nearest Neighbors:* We considered two different distance measures (Euclidean or Manhattan) and the following values for the number of neighbors $k \in \{3, 5, 11, 19\}$.
- *Multi-layer Perceptron (MLP):* We varied the size of the hidden layer in $\{2, 3, 5\}$, the number of epochs in $\{50, 100\}$, and the l_2 regularization parameter in $\{0.001, 0.01, 0.1\}$.
- *Support Vector Regression:* We varied the trade-off parameter $C \in \{1, 10, 100, 1000\}$, and the kernel width $\gamma \in \{0.001, 0.01, 0.1\}$.

Regarding model implementation, all models were trained using Python and the scikit-learn library. The NLP variables related to sentiment analysis were constructed using a pre-trained multilingual sentiment analysis pipeline available from the Hugging Face Transformers library.[4] We applied these models in a Zero-Shot capacity to show their utility without the expensive overhead of domain-specific fine-tuning. This BERT-based model is fine-tuned on multilingual product reviews to predict a 1-5 star rating. The output was then mapped to categorical labels: Negative (for 1 or 2 stars), Positive (for 4 or 5 stars), and Neutral (for 3 stars). This approach captures the general polarity of the text effectively, leveraging the model's understanding of Spanish semantics.

The NER variables were obtained using the spaCy library. We loaded spaCy's large Spanish language model (es-core-news-lg), which is pre-trained specifically on news and web text, making it highly suitable for analyzing social media posts from a media outlet. We specifically extracted the PER (Person) tag to create the NLP.NER feature. This automated feature serves as a counterpoint to the manually curated T.Personalities feature, allowing us to see if any person mention drives engagement, or only specific stars. Prior to entity extraction, a custom exclusion list was applied to handle domain-specific abbreviations and terms. This NER model was also used without further fine-tuning on our specific dataset. Finally, the tailored variables referring to TV presenters were manually coded by exploring different variants of their names.

4.2 Results summary

A summary of the results is presented in Table 1, which considers the best regression performance for various techniques. The performance is assessed in terms of RMSE, MAPE, and R^2 for the two social media platforms. The smallest RMSE and MAPE, and the largest R^2, are highlighted in bold for each platform.

In Table 1, we observe that XGBoost is among the top-performing models for both media platforms. This result shows that of ensemble methods and machine learning in general tend to perform better on average compared to traditional regression approaches based on linear functions or trees.

Another important conclusion from Table 1 is that our framework leads to positive predictive results in general, confirming RQ1. However, the performance is not similar across both social media platforms: engagement models are significantly more accurate for Instagram than for Facebook.

To further analyze the performance of XGBoost, Figs. 4-5 present a sensitivity analysis for this method when considering different hyperparameter values. These figures illustrate the validation performance in terms of R^2 for different combinations of learning rate, number of trees, and maximum depth of the tree as a pruning strategy.

Figs. 4, 5 show relatively stable performance for the various hyperparameter combinations, with some exceptions. Best performance is usually achieved with fewer estimators (100) and with pruned trees (maximum depth of 3). Despite the stability in the results, we suggest following the proposed methodology and evaluating different hyperparameter combinations using an adequate validation approach instead of using default configurations.

[4] nlptown/bert-base-multilingual-uncased-sentiment - Hugging Face, accessed: october 23, 2025, https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment.

Table 1 Predictive performance summary for the ten regression methods, the three evaluation measures (RMSE, MAPE, and R^2), and the two social media platforms (Facebook and Instagram)

	Facebook			Instagram		
Method	RMSE	MAPE	R^2	RMSE	MAPE	R^2
OLS Reg	1.603	28.06	0.12	0.917	19.67	0.67
Ridge Reg	1.568	27.13	0.10	0.933	21.17	0.70
LASSO Reg	1.568	26.87	0.10	0.906	22.08	0.71
Dec. Tree	1.649	28.23	0.082	1.114	19.57	0.60
R. Forest	**1.487**	25.40	0.13	0.806	20.42	0.79
XGBoost	**1.487**	**25.14**	**0.17**	**0.787**	17.39	**0.82**
k-NN	1.500	25.56	0.072	0.933	17.37	0.69
MLP	1.507	26.34	0.085	0.943	22.08	0.71
SVR	1.603	26.08	0.078	0.849	**16.56**	0.80

The best performance is highlighted in bold for each dataset and evaluation measure. The training and test sample sizes in the out-of-time validation strategy are 2358 and 567 for Facebook, and 508 and 123 for Instagram, respectively

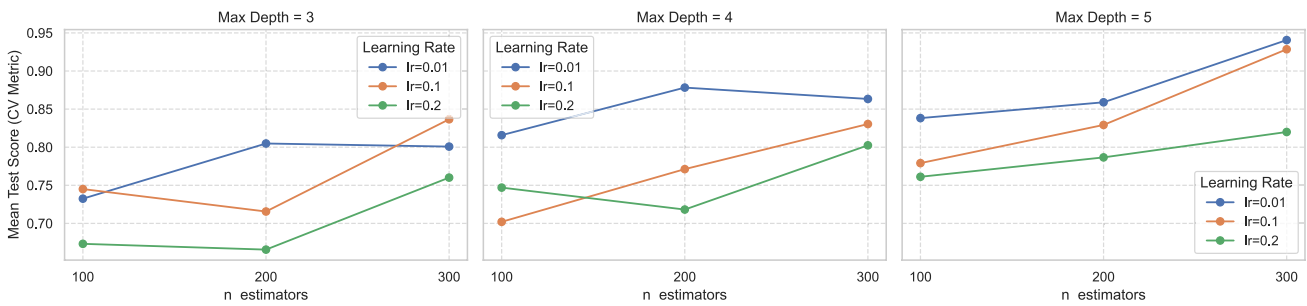


Fig. 4 Influence of the hyperparameters in terms of validation performance (R^2) for the best-performing regression method (XGBoost). Facebook platform

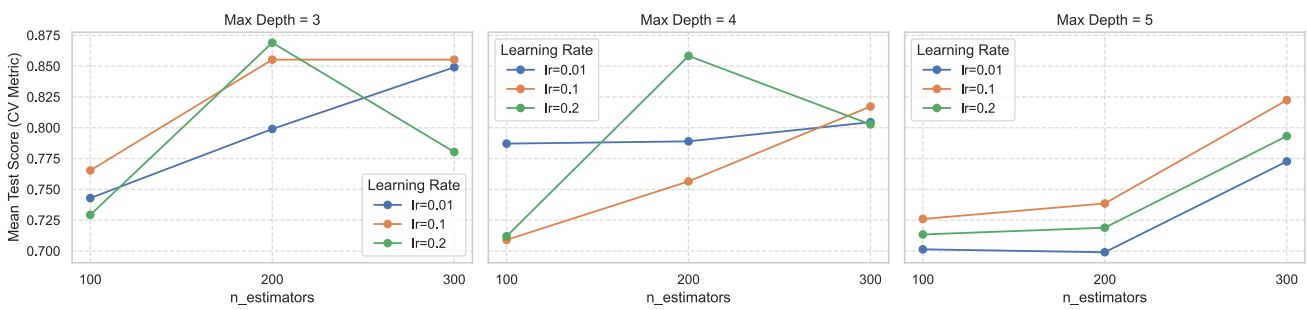


Fig. 5 Influence of the hyperparameters in terms of validation performance (R^2) for the best-performing regression method (XGBoost). Instagram platform

Next, we perform an ablation study to assess the contribution of different feature groups and discuss our research questions RQ2 and RQ3 using the best-performing model, XGBoost. Table 2 presents the regression performance for this method in terms of the RMSE, MAPE, and R^2 for the two social media platforms under four different experimental settings (EXP) (see a detailed description of the data sources in Sect. 3.1):

- EXP1: Structural variables only (S.*).
- EXP2: Structural variables and tailored variables (S.*+T.*).
- EXP3: Structural variables and NLP variables (S.*+NLP.*).
- EXP4: Structural variables and features related to persons in general (S.*+T.Personalities+ NLP.NER).

The smallest RMSE and MAPE, and the largest R^2, are highlighted in bold for each platform. We also include in Table 2 the performance of XGBoost with all the variables for illustrative purposes. We refer to this experimental setting as EXP0.

In Table 2, we observe that adding Tailored variables (EXP2) provided a massive boost to performance, almost doubling the explained variance on Facebook compared to

Table 2 Predictive performance for the XGBoost method under different experimental settings

	Facebook			Instagram		
Exp. Setting	RMSE	MAPE	R^2	RMSE	MAPE	R^2
EXP0 (all)	2.21	25.37	0.17	0.66	18.56	0.78
EXP1 (S.*)	2.42	26.08	0.09	1.36	24.30	0.67
EXP2 (S.*+T.*)	2.24	25.24	0.15	0.7	17.71	0.76
EXP3 (S.*+NLP.*)	2.25	26.16	0.08	1.31	25.63	0.69
EXP4 (S.*+T.Pers.+ NLP.NER)	2.31	25.73	0.12	0.72	20.49	0.75

The three evaluation measures (RMSE, MAPE, and R^2) and the two social media platforms (Facebook and Instagram) are considered in this analysis

Table 3 Coefficients of the multinomial linear regression obtained with data from the Facebook social media platform

Variable	β	SE(β)	t	$P > \|t\|$	[0.025	0.975]
Offset	4.7354	0.169	28.101	0.000	4.405	5.066
S.Content.Hashtag	−1.8198	0.186	−9.775	0.000	−2.185	−1.455
S.Content.Uppercase-Word	0.6374	0.091	6.967	0.000	0.458	0.817
S.Content.Question-Symbol	−0.8390	0.093	−9.025	0.000	−1.021	−0.657
S.Type.Video	0.4454	0.162	2.752	0.006	0.128	0.763
T.Personalities.Presenter-M	1.3492	0.277	4.863	0.000	0.805	1.893
T.Personalities.Presenter-F	0.5698	0.205	2.780	0.005	0.168	0.972
T.Program-Name	1.3781	0.190	7.253	0.000	1.006	1.751
T.Rating	0.2021	0.041	4.895	0.000	0.121	0.283
T.BTTP.Day-Sun	−0.7286	0.256	−2.851	0.004	−1.230	−0.228

Table 4 Coefficients of the multinomial linear regression obtained with data from the Instagram social media platform

Variable	β	σ_β	t	$P > \|t\|$	[0.025	0.975]
Offset	1.8201	0.389	4.684	0.000	1.057	2.584
S.Type.Photo	−1.1403	0.178	−6.400	0.000	−1.490	−0.790
S.Type.Video	−1.8970	0.159	−11.906	0.000	−2.210	−1.584
S.Type.Post	2.8265	0.257	11.018	0.000	2.322	3.331
S.Type.Reel	4.2172	0.273	15.475	0.000	3.682	4.753
S.Length	−0.0390	0.237	−0.164	0.869	−0.505	0.427
S.Content.Exclamation-Symbol	0.5381	0.136	3.955	0.000	0.271	0.805
T.Personalities.Presenter-M	0.7399	0.155	4.767	0.000	0.435	1.045
T.Personalities.Presenter-F	0.9480	0.141	6.719	0.000	0.671	1.225
T.Rating	0.4114	0.045	9.224	0.000	0.324	0.499
T.BTTP.Day-Sun	−0.0389	0.246	−0.158	0.875	−0.523	0.445
T.BTTP.Day-Thurs	−0.2948	0.090	−3.281	0.001	−0.471	−0.118
NLP.NER-Person	−0.2460	0.083	−2.969	0.003	−0.409	−0.083
NLP.Sentimient-Pos	−0.1131	0.077	−1.478	0.140	−0.263	0.037

structural features alone. This confirms that knowing who is in the post (specific presenters) and when it was posted (TV rating context) is far more valuable than knowing the general sentiment. Furthermore, while NLP alone (EXP3) added little value over structural features, the combination of all features (EXP0) yielded the highest results. This suggests that NLP features are not silver bullets but are valuable modifiers: a positive sentiment might boost a presenter post, even if it does not save a generic post. We conclude that both RQ2 and RQ3 can be considered true, emphasizing the importance of TV personalities over other tailored variables such as BTTP and NLP variables such as post polarity.

4.3 Explainability and decision-making

Next, we analyze the interpretable models to further discuss the influence of the various data sources. First, the coefficients for the multinomial linear regression are presented in Tables 3 and 4 for the Facebook and Instagram platforms, respectively. These tables report the variable coefficients β, their standard deviations SE(β), the t-statistics, the p-values $P > |t|$, and the confidence intervals for each β. As indicated in the previous section, a backward elimination approach was performed to include only significant covariates. The variables were scaled between 0 and 1.

It can be observed in Table 3 that only structural (S.*) and tailored (T.*) variables appear relevant for the Facebook model. Regarding the first source, the inclusion of videos and uppercase words improves engagement, in contrast to question marks and hashtags. For the second source, the rating and the mention of both the male and female presenters in the posts leads to more viewer engagement. The same occurs when the name of the program is mentioned in the post. Furthermore, there is a positive correlation with the rating of the last program and the success of a post, suggesting that it is a good idea to generate posts on days of high ratings. Finally, posts tend to be less successful on Sunday.

In contrast, for the Instagram platform (see Table 4), all feature sources appear relevant; the final model includes NLP variables such as positive polarity and NER. We found that Reels, posts, presenter mentions, the rating, and the use of exclamation symbols positively influence engagement. Conversely, photos and videos (as publication types), post length, and posts made on Thursday or Sunday are associated with lower engagement. This suggests that for Instagram, the post type is a highly relevant driver of engagement, seemingly more so than other structural variables. As with Facebook, tailored variables (such as presenter mentions, the rating, and the posting day) are also significant. These results confirm both RQ2 and RQ3.

The resulting decision trees are presented in Appendix B. In Fig. 8, we observe that the main variables related to the first two levels of the tree are the two TV presenters and the use of question marks in the post. In the third level, other structural and tailored variables appear relevant, such as the rating, the use of videos, and uppercase words. For Instagram (Fig. 9), the same tailored variables are relevant in the first three levels: the two presenters and the rating. However, the relevant structural variables are the use of emojis and the length of the post. Note that the results obtained with both the linear regressions and the decision trees are consistent, conferring validity to our approach.

Note that neither linear regression nor the decision trees were implemented because XGBoost was the best-performing approach on average. However, these traditional techniques achieved competitive results and can be considered for interpretability purposes. Decision trees achieved the worst performance among the various predictors, and therefore the conclusions drawn from this method should be considered with caution. Next, the most relevant variables obtained with Boruta-SHAP are presented in Figs. 6 and 7 for the Facebook and Instagram platforms, respectively. XGBoost is utilized as the baseline regression method.

Similar to the results obtained for the interpretable methods, the eight most valuable features come from different sources. The two presenters appear as relevant for both platforms, along with the average rating of the last broadcast of the TV show. The Facebook model is strongly influenced by structural variables such as uppercase words, hashtags, or question marks, while the Instagram model relies more on the type of post, such as reels, videos, images, or standard posts. The mention of a person appears relevant for the Facebook model, being the only NLP variable among the eight most relevant ones. For the Instagram model, the presence of emojis and the length of the post are relevant. This shows that the importance of structural variables strongly depends on the social media platform.

Synthesizing these findings provides valuable qualitative insights into the specific cues driving engagement differently across platforms. On Facebook, engagement appears more strongly influenced by visual cues like Videos and linguistic choices that convey emphasis (Uppercase Words) rather than interactive prompts (Question Marks) or categorization

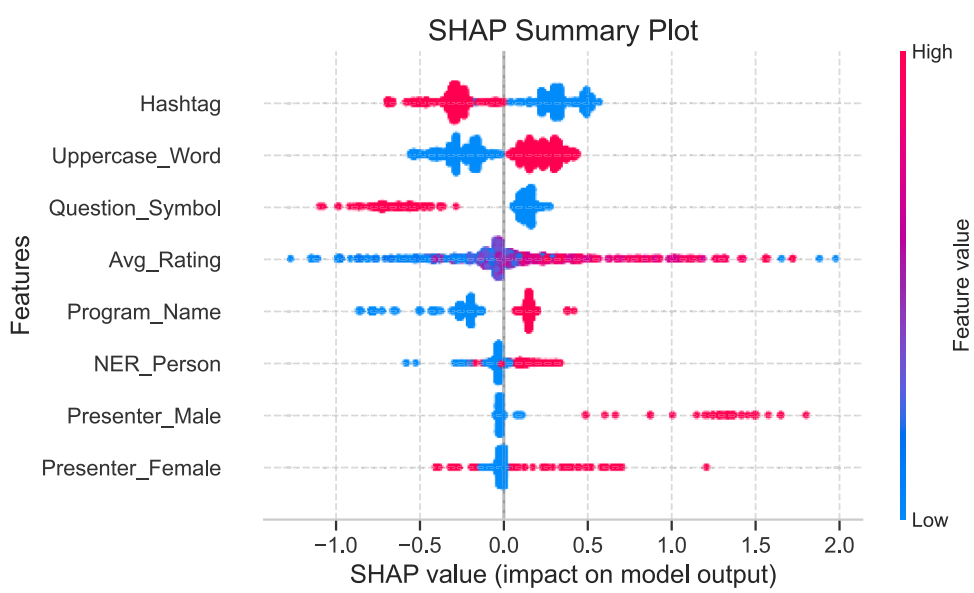


Fig. 6 Graphical representation of the Boruta-SHAP method, indicating the most relevant variables of the best-performing regression approach (XGBoost) for the Facebook social media platform

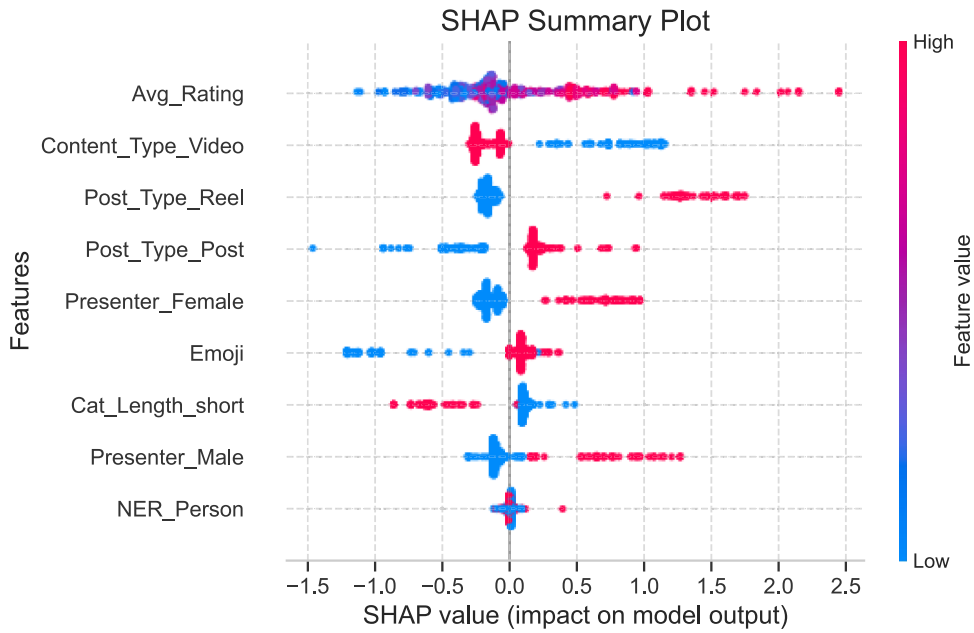


Fig. 7 Graphical representation of the Boruta-SHAP method, indicating the most relevant variables of the best-performing regression approach (XGBoost) for the Instagram social media platform

(Hashtags), which showed negative correlations. The contextual cue of mentioning specific TV presenters or the program name itself is a powerful positive driver, alongside aligning posts with high TV ratings. This suggests Facebook audiences respond well to direct, emphatic content related to core programming and personalities, aligning with its nature supporting diverse interactions including discussions.

Conversely, Instagram engagement seems exceptionally sensitive to format cues, with Reels and standard Posts significantly outperforming other types. Visual cues like Emojis and linguistic factors such as brevity (short post length) and enthusiasm (Exclamation Marks) also play a notable role. While mentions of presenters remain crucial, the linear regression results surprisingly showed negative coefficients for general person mentions (NLP.NER-Person) and positive sentiment. While seemingly counter-intuitive and potentially an artifact of the specific linear model, this might suggest that simple positivity or mentioning any person is less impactful than the highly specific context of the main presenters or the visually-driven format choice on this platform. Overall, the analysis highlights that optimizing for engagement requires a platform-specific understanding of which visual, linguistic, and contextual cues resonate most effectively with each audience.

While the main Boruta-SHAP analysis (as shown in Figs. 6, 7) is effective for identifying the most important features, it does not fully reveal how these features interact. To explore variable interdependencies learned by the model, we further employ SHAP interaction analysis, as suggested by Lundberg et al. (2020). This method relies on SHAP interaction values, which are an extension of the SHAP framework. For any two features, i and j, the SHAP values can be decomposed such that the total effect includes a main effect and an interaction effect (ϕ_{ij}). We visualize these interdependencies using SHAP interaction dependence plots. This analysis is presented in Appendix C, and features two illustrative examples (one for each social network; see Figs. 10 and 11).

5 Discussion and managerial implications

The results of this study transcend academic metrics, offering a concrete playbook for media managers navigating the complexities of the attention economy. For community managers, the findings dictate a clear operational shift: the practice of cross-posting must end, as our models show that Facebook and Instagram function as distinct ecosystems requiring tailored strategies. On Facebook, the objective should be optimizing for conversation.

Managers are encouraged to use high-frequency content featuring questions and emphatic capitalization to trigger the platform's comment-weighting logic. Conversely, hashtags should be used sparingly or eliminated entirely on Facebook, as linear models indicate a negative coefficient, suggesting they act as "exit signs" that lead users away from the content. Instagram, by contrast, demands an optimization for format rather than text; the caption is secondary, and the platform should be treated as a visual magazine rather than a link repository. Specifically, content not formatted as a Reel is likely to remain invisible. Furthermore, while emojis show a positive correlation with engagement on Instagram, helping to signal tone, they remain neutral on Facebook and should not consume valuable character counts in

traffic-driving posts.For executive producers, these findings offer a tactical advantage rooted in the synchronization of media assets.

The strong predictive power of television ratings (*T.Rating*) confirms the "second screen" phenomenon, where social audiences are primed when the broadcast show peaks. Consequently, producers should synchronize high-effort social posts to occur immediately after high-rating TV segments to capitalize on this "halo effect," rather than scheduling top-tier content for off-hours. Additionally, the specific influence of variables such as "Presenter-M" versus "Presenter-F" enables data-driven talent management. By monitoring specific social engagement scores, visible via SHAP monitoring, producers can identify early warning signs of waning audience affinity, which may eventually impact television ratings, allowing for proactive adjustments in programming and talent utilization.

Finally, we must acknowledge a significant limitation often cited in recent literature: this study relies on unimodal content analysis, processing metadata (e.g., "Is it a photo?") rather than the raw pixel data. Current state-of-the-art research utilizes Multimodal Deep Learning, employing Convolutional Neural Networks to extract visual features such as color temperature or facial expressions and fusing them with text embeddings. While these methods can achieve higher accuracy, they are computationally intensive and lack the transparency of our metadata-based approach. However, the high R^2 of 0.82 achieved on Instagram suggests that, within the specific domain of TV analytics, metadata serves as a robust proxy for content. Variables such as "Reel" and "Presenter" capture the vast majority of the variance, potentially rendering pixel-level analysis a case of diminishing returns for operational environments. Future research should aim to integrate lightweight visual embeddings, such as CLIP, to determine if the remaining 18% of variance can be explained by visual aesthetics without sacrificing efficiency.

6 Conclusions

In this paper, we present a novel framework for analyzing social media platforms for a TV channel using text analytics. The goal is to predict the success of a post while understanding the main drivers of viewer engagement. This research proposes an integrated approach based on variables that characterize the posts, the definition of tailored variables for TV programs, and the automatic construction of features utilizing NLP techniques.

Central to our analysis was the hypothesis that traditional and machine learning methods can accurately predict viewer engagement (RQ1). Through extensive experimentation on two social media platforms, we confirmed this hypothesis. In particular, XGBoost showed a remarkable ability to model the problem, achieving the lowest RMSE on both platforms. Predictive results for Instagram are significantly better than for Facebook, showing that the ability to predict engagement strongly depends on the social media platform.

Our study also focuses on analyzing the influence of TV figures (RQ2) and automatically-generated NLP variables (RQ3) on the success of posts. Our results show that the mention of current TV presenters of the program strongly influences viewer engagement, confirming the importance of tailored variables.

This framework is also an important tool for decision-making: as discussed in Sect. 3.4, the proposed framework goes beyond operational decisions regarding post design and allows TV programs to make adjustments in the TV personalities or the content of the various sections that encompass the program. The XAI strategies considered in this study, such as the analysis of interpretable regression methods and Boruta-SHAP for feature contribution, allow hypothesis testing to gain valuable insights from viewer responses to posts.

Despite the promising results, this study has several limitations that must be critically discussed. First, the dataset is sourced from a single major TV channel in Chile. The specific drivers of engagement identified here may be unique to the Chilean cultural context, the channel's specific brand identity, and the morning talk show genre. It is unlikely that these exact findings would replicate for a news channel in Europe or a sports broadcaster in North America. The study's value, therefore, lies more in the methodological framework itself than in the specific feature importances discovered.

Second, the analysis represents a snapshot in time. Social media is a highly dynamic environment, and the factors that drive engagement today may be irrelevant tomorrow. This leads to the critical issue of model decay. The relevance and predictive power of the trained models are expected to diminish over time, a concept analogous to time decay in attribution modeling where the influence of past events fades. This is particularly true for transient features like *T.Personalities*; a change in the program's hosts would instantly invalidate a significant portion of the model's learned patterns. This limitation underscores the necessity of not viewing the model as a one-time solution but as a dynamic system that requires continuous monitoring and periodic retraining to maintain its validity and utility as a decision support tool.

The limitations of this study open several important avenues for future research that could significantly advance the field of media analytics. The most critical area for improvement is the development of a more sophisticated, weighted engagement metric. As discussed, the simple sum

of interactions is a crude proxy for true engagement. Future work should focus on creating a composite engagement score where different interactions are weighted according to their value. For example, methods from multi-criteria decision-making could be used to assign higher weights to shares and comments than to likes, creating a more meaningful and less noisy target variable that better reflects deep audience involvement.

Another vital direction is to formally test the framework's generalizability via external validation. Sourced from a single Chilean TV channel, the present data inherently carries the potential for cultural bias; the observed engagement patterns may reflect Chilean media consumption habits or cultural norms rather than universal behaviors. Similarly, linguistic bias could be present, as the NLP models might be influenced by specific nuances of Chilean Spanish, potentially limiting the applicability of findings related to linguistic features in other Spanish-speaking regions or different languages altogether.

Therefore, the proposed external validation must involve a large-scale study employing explicit dataset diversity strategies. This would entail applying the methodology to data meticulously collected from multiple TV channels across different countries, ensuring representation from diverse cultural contexts. The dataset must also span various program genres (e.g., news, sports, drama, reality TV) to understand genre-specific engagement dynamics. Incorporating cross-linguistic data would further strengthen generalizability, necessitating the use of robust cross-lingual analytical techniques or model adaptations (see, e.g., Li et al. 2024). Employing appropriate cross-cultural research methodologies throughout this process will be essential. Such a comprehensive, diverse study is critical not only to mitigate the potential cultural and linguistic biases inherent in single-source datasets but also to identify which engagement drivers identified in our study are universal and which are context-specific. This endeavor would lead to a more robust, nuanced, and widely applicable theory of social media engagement in the television industry.

Finally, future research could shift the unit of analysis from the post to the user. Developing user-based engagement models would allow for the segmentation of audiences and the personalization of content. This would enable TV channels to move beyond optimizing individual posts for a general audience and toward tailoring content strategies for specific viewer segments, a common goal in advanced social media analytics.

Appendix A: Comprehensive Feature Engineering Taxonomy

Table 5 Taxonomy of the feature engineering framework, categorizing variables into Structural, Tailored, and NLP-derived groups, detailing their specific preprocessing logic (e.g., One-Hot Encoding, Binary, Count) and theoretical rationale for predicting social media engagement

Category	Variable	Description & Logic	Rationale
Structural (S)	`S.Type`	Categorical: Photo, Video, Reel, Status. (One-Hot Encoded)	Platform algorithms prioritize specific formats (e.g., Reels on IG)
	`S.Length`	Numerical: Character count of th caption (Normalized)	Brevity vs. Detail trade-offs vary by platform strategy
	`S.Hashtag`	Count: Number of hashtags used	Discoverability vs. Spam appearance
	`S.Emoji`	Binary & Count: Presence (0/1) and total count	Indicators of informality and emotional tone
	`S.Caps`	Binary: Presence of words in ALL CAPS	Emphasis, shouting, or breaking news urgency
	`S.Punct`	Binary: Presence of '?' or '!'	Calls to action (Questions) vs. Emotional outbursts
Tailored (T)	`T.Pers`	Binary: Specific mentions of the show's main hosts	Tests the "Star Power" and parasocial attachment hypothesis
	`T.Program`	Binary: Mention of the show's official name	Branding consistency
	`T.Rating`	Numerical: TV rating of the preceding broadcast	Captures the "Second Screen" momentum effect
	`T.BTTP`	Categorical: Day of Week and Time of Day	Aligns with user activity cycles (Best Time To Post)
NLP	`NLP.Sent`	Categorical: Positive, Neutral, Negative (BERT)	Automated emotional valence
	`NLP.NER`	Binary: Presence of entity tag `PER` (spaCy)	Distinguishes between "posts about people" vs. "posts about things."

Appendix B: Decision trees for the two social media platforms

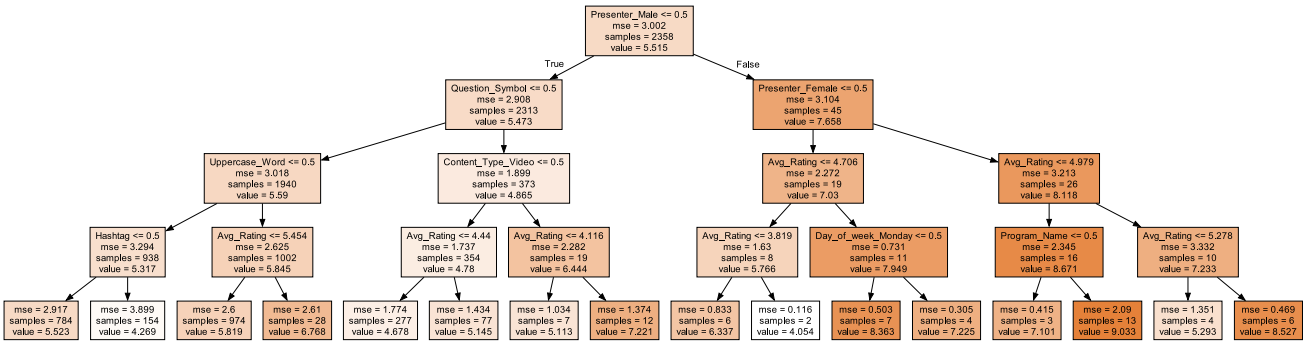


Fig. 8 Decision tree for the Facebook platform

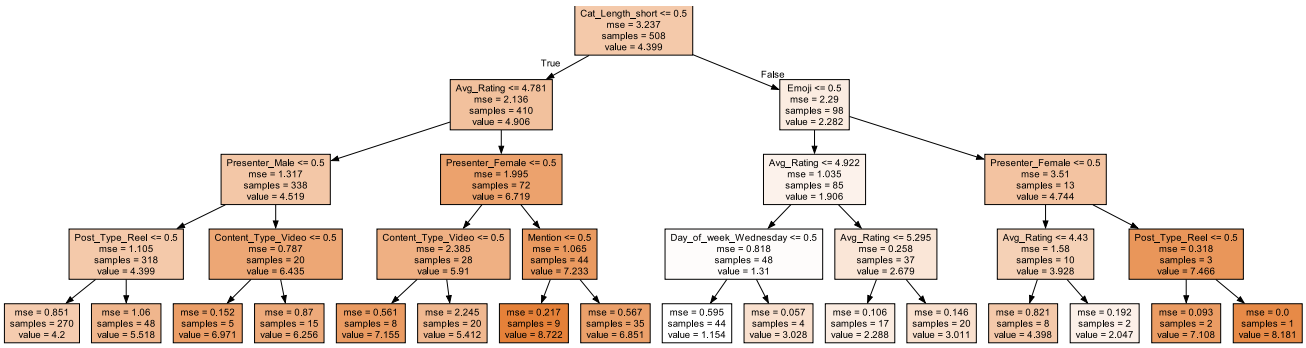


Fig. 9 Decision tree for the Instagram platform

Appendix C: SHAP interaction analysis

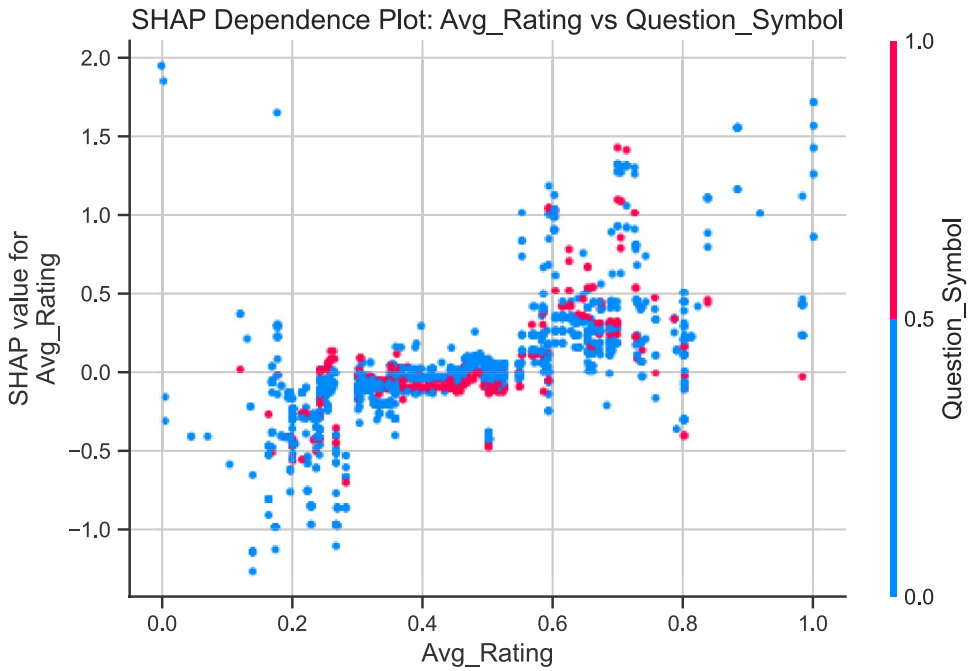


Fig. 10 Boruta-SHAP interaction plot for the Facebook social media platform. The x-axis represents the value of a main feature (Average Rating in this case). The y-axis represents the SHAP value of that same feature, indicating its impact on the model's output. Each point on the plot is colored by the value of a second, interacting feature (Question Symbol in this case). A clear vertical separation of colors in the plot (e.g., the cluster of red dots at the middle of the plot) provides strong evidence of an interaction. It visually shows how the marginal contribution of the main feature (Average Rating) changes based on the value of the secondary feature (Question Symbol), thus revealing synergistic effects the model has captured

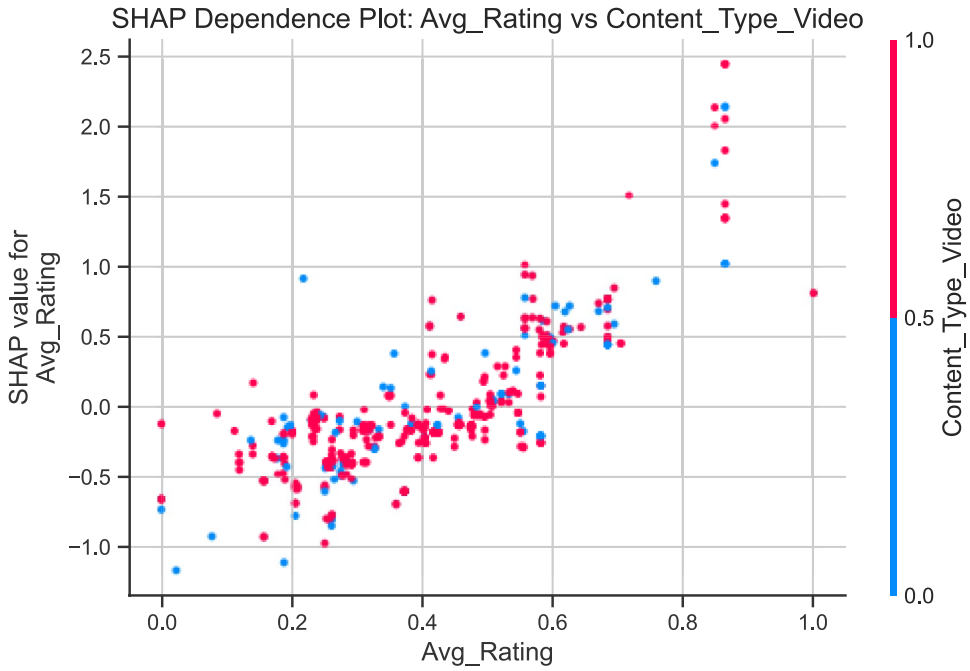


Fig. 11 Boruta-SHAP interaction plot for the Instagram platform. Average Rating is the main variable (x-axis), colored by the interacting feature Post Type = Video. In this case, no evidence of interdependence between these two variables is observed

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Conceptualization, Methodology, Supervision, Writing- Original draft preparation.

Data availability Due to sensitivity concerns and restrictions from the providing company, the dataset is not openly accessible. However, an anonymized sample and the codes used for the analysis can be obtained from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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